

GAFSA, Tunisia

WMO: 607450

Lat: 34.422N

Lon: 8.823E

Elev: 1060

StdP: 14.14

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.5	37.7	20.9	16.3	55.4	23.4	18.3	53.7	25.2	53.3	22.3	55.1	6.2	60	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	107.6	68.5	104.2	68.3	101.0	68.1	73.5	91.4	72.3	90.5	71.2	89.7	10.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.1	111.3	79.4	67.7	105.8	78.5	66.0	99.8	78.4	37.7	91.2	36.6	90.3	35.6	89.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	24.0	21.2	18.8	DB	29.7	112.2	2.4	1.9	27.9	113.5	26.5	114.6	25.2	115.6	23.4	117.0
(5)				WB	27.6	76.0	2.4	1.9	25.9	77.4	24.5	78.5	23.1	79.6	21.4	80.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.6	50.3	52.7	59.0	65.9	74.0	82.4	87.7	87.1	79.8	71.3	59.7	52.2	(6)
(7)		DBStd	14.18	3.86	4.92	5.79	5.59	5.84	5.77	4.44	4.28	5.46	5.44	5.34	4.50	(7)
(8)		HDD50	101	43	23	4	0	0	0	0	0	0	0	2	29	(8)
(9)		HDD65	1639	455	346	201	53	3	0	0	0	0	9	174	398	(9)
(10)		CDD50	6890	53	97	284	478	745	972	1168	1150	893	660	293	97	(10)
(11)		CDD65	2953	0	1	16	81	283	522	703	685	443	204	15	0	(11)
(12)		CDH74	37619	0	14	231	900	3105	6829	10286	9680	4811	1666	96	1	(12)
(13)		CDH80	21385	0	2	54	291	1429	4031	6569	6000	2412	588	9	0	(13)
(14)	Wind	WSAvg	9.4	8.2	9.2	10.1	11.0	11.0	10.8	9.7	9.4	9.3	8.4	8.0	7.9	(14)
(15)	Precipitation	PrecAvg	6.4	0.6	0.5	0.7	0.6	0.5	0.2	0.1	0.2	0.7	0.8	0.6	0.7	(15)
(16)		PrecMax	11.5	3.9	2.5	3.9	3.2	2.4	1.2	0.6	1.7	4.1	3.7	2.9	7.2	(16)
(17)		PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.4	0.8	0.6	0.7	0.7	0.6	0.3	0.1	0.4	0.8	0.9	0.7	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.2	75.5	86.1	92.9	102.1	109.2	111.5	109.7	103.8	94.6	80.7	70.0	(19)
MCWB			52.3	54.5	57.7	61.3	65.2	67.1	68.7	69.5	68.2	64.6	61.2	54.2	(20)	
2%		DB	64.7	70.6	79.9	87.3	95.4	104.4	107.9	106.2	98.8	90.1	76.8	66.5	(21)	
		MCWB	50.9	53.0	56.5	59.8	64.0	66.8	68.9	69.1	68.2	64.6	59.0	53.3	(22)	
5%		DB	62.4	67.3	75.4	82.8	91.7	100.4	104.4	103.1	95.2	86.2	73.4	64.3	(23)	
		MCWB	50.3	51.5	55.2	59.1	63.3	66.5	68.6	68.8	68.4	64.3	58.0	52.3	(24)	
10%		DB	59.7	64.0	71.7	78.9	87.9	96.7	101.1	100.2	91.7	82.6	70.2	61.6	(25)	
		MCWB	49.0	50.3	54.2	57.7	62.5	66.0	68.3	68.8	68.2	63.7	56.8	51.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	56.9	61.2	64.4	69.0	71.6	74.8	75.6	75.2	71.5	64.0	57.6	(27)
MCD B			63.1	70.4	73.7	85.2	87.7	90.2	92.7	93.2	89.4	82.3	75.5	64.4	(28)	
2%		WB	52.8	54.4	58.5	62.0	67.1	69.9	72.8	73.7	72.8	69.2	61.7	55.6	(29)	
		MCD B	61.2	67.6	74.9	79.9	87.3	91.8	94.8	91.3	87.1	79.9	72.5	62.6	(30)	
5%		WB	51.3	52.6	56.9	60.4	65.5	68.7	71.5	72.4	71.5	67.5	59.9	54.0	(31)	
		MCD B	59.7	64.4	72.2	77.2	85.2	91.6	93.9	90.8	85.3	79.1	69.8	61.2	(32)	
10%		WB	50.0	50.9	55.2	59.1	64.0	67.5	70.4	71.4	70.2	66.0	58.2	52.3	(33)	
		MCD B	58.3	61.8	68.9	75.5	83.0	90.8	92.6	90.4	84.4	77.7	67.8	60.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.4	20.4	22.0	22.6	23.7	25.3	26.0	24.5	21.2	20.4	19.5	18.5	(35)
MCD BR			22.9	25.4	27.6	28.8	28.0	29.0	29.8	28.2	25.6	24.9	22.5	21.6	(36)	
MCWBR			12.7	12.8	11.7	10.4	9.3	8.7	8.8	8.6	8.0	9.2	10.4	11.9	(37)	
5% WB		MCD BR	20.1	22.2	23.9	24.6	24.7	26.1	26.6	24.4	22.0	20.1	19.5	18.4	(38)	
		MCWBR	11.9	11.8	10.6	9.5	8.7	8.5	8.5	8.0	7.6	8.3	10.0	10.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.360	0.446	0.501	0.532	0.504	0.536	0.519	0.505	0.453	0.390	0.343	(40)	
taud		2.313	2.216	1.952	1.836	1.793	1.883	1.822	1.862	1.902	2.000	2.138	2.276	(41)		
Ebn at Noon		274	278	260	252	246	252	243	244	240	241	247	261	(42)		
Edh at Noon		32	39	56	66	69	63	67	63	58	48	38	31	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1001	1297	1662	2013	2259	2485	2479	2201	1781	1392	1062	899	(44)	
(45)		RadStd	57	56	74	90	109	82	59	69	69	60	41	58	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	+0.40	N/A	N/A	N/A	-0.59	N/A	N/A	N/A	+185	+108	

Nomenclature: See separate page